

Predictive Modeling and Feature Impact Analysis

for Hypertension Diagnosis

A Machine Learning Approach to Early Risk Prediction Using Clinical and Lifestyle Data

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Abstract

Hypertension remains one of the most significant modifiable risk factors for cardiovascular disease worldwide, yet it is frequently under-diagnosed until complications arise. This project investigates whether routinely collected demographic and lifestyle data can be used to build an accurate, interpretable classifier for hypertension risk. Three supervised learning algorithms — Logistic Regression, K-Nearest Neighbors (KNN), and Random Forest — were trained and evaluated on a clinical dataset of approximately 1,900 patient records. Beyond standard model comparison, this work places particular emphasis on **feature impact analysis**: how model accuracy and generalization change as features are incrementally added or removed. The Random Forest classifier achieved the strongest performance (93.3% accuracy), followed by Logistic Regression (87.8%) and KNN (77.7%). Feature sensitivity experiments revealed a clear "sweet spot" of seven features beyond which distance-based and linear models degrade due to the curse of dimensionality and multicollinearity, while the ensemble-based Random Forest remained stable. These findings underscore that in clinical machine learning contexts, model choice and disciplined feature selection are as important as raw data volume.

1. Introduction

1.1 Background

Hypertension, or persistently elevated blood pressure, affects an estimated 1.3 billion adults globally and is a leading contributor to stroke, heart attack, kidney disease, and premature death. Because hypertension is frequently asymptomatic in its early stages, it is often referred to as a "silent killer" — many patients are unaware of their condition until a routine check-up or an acute cardiovascular event. Early identification of at-risk individuals through predictive screening tools can enable timely lifestyle intervention and treatment, reducing long-term healthcare burden. Machine learning offers a scalable way to flag at-risk patients using data that is already collected in most clinical intake workflows, without requiring additional invasive testing.

1.2 Problem Statement

The central challenge addressed in this project is twofold. First, can hypertension status be predicted with clinically useful accuracy using only demographic and lifestyle indicators — such as age, salt intake, stress score, sleep duration, BMI, and family history — rather than invasive laboratory measurements? Second, how many of these features are actually necessary? Clinical datasets often contain redundant or weakly informative variables, and naively including all available features can introduce noise, inflate model complexity, and paradoxically reduce predictive performance, particularly for algorithms sensitive to high-dimensional feature spaces. This project treats feature selection not as a preprocessing afterthought, but as a primary object of empirical study.

1.3 Project Objectives

- Preprocess and encode a real-world clinical hypertension dataset, handling missing values, outliers, and categorical variables.
 - Implement and rigorously compare three classification algorithms — Logistic Regression, K-Nearest Neighbors, and Random Forest — under identical train/test conditions.
 - Empirically quantify how incrementally adding and removing features affects model accuracy and generalization, in order to identify an optimal, minimal feature subset.
 - Translate model coefficients and feature importances into interpretable, clinically relevant insights.
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2. Dataset Overview and Description

2.1 Data Source

The analysis is built on **hypertension_dataset.csv**, a structured tabular dataset of patient-level records combining demographic attributes, lifestyle behaviors, and clinical history. Each row represents a single patient, and the dataset was used in its entirety for the preprocessing, modeling, and feature impact experiments described in the following sections.

2.2 Feature Dictionary

The dataset contains the following ten predictor columns, spanning demographic, behavioral, and clinical history information:

Feature	Type	Description
Age	Numeric	Patient age in years.
Salt_Intake	Numeric	Estimated daily dietary salt intake.
Stress_Score	Numeric	Self-reported psychological stress rating.
BP_History	Categorical	Prior blood-pressure classification/history (multi-class).
Sleep_Duration	Numeric	Average hours of sleep per night.
BMI	Numeric	Body Mass Index.
Medication	Categorical	Current medication category (multi-class).
Family_History	Binary	Family history of hypertension (Yes/No).

Exercise_Level	Categorical	Self-reported physical activity level (multi-class).
Smoking_Status	Binary	Smoker / Non-Smoker.

2.3 Target Variable

The prediction target is **Has_Hypertension**, a binary label (Yes/No, encoded as 1/0) indicating whether the patient has been diagnosed with hypertension. All three models were trained as binary classifiers against this label.

2.4 Data Quality

Prior to modeling, the dataset was profiled for missing values, data type consistency, and class balance. The target classes were moderately imbalanced but not severely so — the test split ultimately contained 101 negative and 137 positive cases (238 total), a ratio close enough to 40/60 that accuracy remained a reasonably reliable metric, supplemented with precision, recall, and F1 score to guard against misleading conclusions from class imbalance.

3. Data Preprocessing and Feature Engineering

3.1 Handling Missing Values

Records with null values in critical fields were identified and removed rather than imputed, given that the proportion of missing records was small relative to the overall dataset size. This conservative strategy avoids introducing synthetic bias into clinically sensitive fields such as BMI or blood pressure history, at the cost of a modest reduction in sample size.

3.2 Outlier Detection and Capping

Continuous variables — particularly **BMI** and **Sleep_Duration** — were screened for extreme values using the Interquartile Range (IQR) method. Values falling outside $1.5 \times \text{IQR}$ from the first and third quartiles were capped (winsorized) rather than dropped, preserving sample size while limiting the influence of implausible or erroneous entries (e.g., a BMI of 70 or a reported sleep duration of 15 hours) that could otherwise skew distance-based models like KNN and destabilize the coefficients of Logistic Regression.

3.3 Categorical Encoding

3.3.1 Binary Encoding

Binary categorical fields — **Family_History** (Yes/No) and **Smoking_Status** (Smoker/Non-Smoker) — were mapped directly to 1/0 integer flags, avoiding unnecessary dimensionality expansion for variables that are inherently two-valued.

3.3.2 One-Hot Encoding

Multi-class categorical variables — **BP_History**, **Medication**, and **Exercise_Level** — were transformed using `pd.get_dummies(drop_first=True)`. Dropping the first category of each variable avoids the dummy-variable trap (perfect multicollinearity between one-hot columns), which is particularly important for

Logistic Regression, whose coefficient estimates and interpretability degrade under collinear inputs.

3.4 Feature Scaling

All numeric features were standardized using `StandardScaler` (zero mean, unit variance) prior to training Logistic Regression and KNN. This step is essential for both algorithms: KNN's distance calculations are dominated by whichever feature has the largest raw numeric scale (e.g., Age vs. a 0–1 binary flag) unless all features are placed on a common scale, and gradient-based optimization for Logistic Regression converges faster and more reliably on standardized inputs. Random Forest, being a tree-based, split-threshold algorithm, is scale-invariant and was trained on the unscaled (but encoded) feature set.

4. Methodology: Classification Algorithms

4.1 Logistic Regression

Logistic Regression was used as the interpretable baseline model. As a linear classifier, it estimates the log-odds of hypertension as a weighted linear combination of the input features, making its coefficients directly interpretable as the direction and relative magnitude of each feature's association with the outcome.

4.2 K-Nearest Neighbors (KNN)

KNN is a non-parametric, instance-based algorithm that classifies a patient according to the majority class among its k nearest neighbors in feature space. It makes no assumptions about the underlying data distribution, but is known to be highly sensitive to irrelevant or noisy features and to suffer from the curse of dimensionality as the feature space grows.

4.3 Random Forest

Random Forest is an ensemble method that aggregates the predictions of many decorrelated decision trees, each trained on a bootstrap sample of the data with a random subset of features considered at each split. This structure makes it inherently robust to outliers, non-linear relationships, and irrelevant features, since individual trees that rely on uninformative splits are diluted in the ensemble average.

4.4 Experimental Setup

All three models were trained and evaluated under identical conditions to ensure a fair comparison: an 80/20 train-test split, with performance measured using Accuracy, Precision, Recall, and F1-Score. This multi-metric approach was chosen deliberately over accuracy alone, since a purely accuracy-driven evaluation can mask poor performance on the minority class in a clinical screening context, where false negatives (missed hypertension cases) carry a materially higher cost than false positives.

5. Experimental Design: Feature Impact Analysis

5.1 Feature Ranking

A baseline Random Forest model was trained on the full feature set and used to rank all features by **Gini Importance** — the average reduction in node impurity attributable to each feature across all trees in the

forest. This ranking served as the ordering scheme for the incremental feature experiments described below, ensuring that the most informative variables were introduced first.

5.2 Incremental Feature Addition Protocol

To measure how each algorithm responds to growing feature-space complexity, a loop was designed that iteratively adds one feature at a time — starting from the single most important feature and expanding up to the full set of N features, in order of the Gini ranking from Section 5.1. At each step, all three models (Logistic Regression, KNN, Random Forest) were retrained from scratch on the current feature subset and evaluated on the held-out test set, producing an accuracy trajectory for each algorithm as a function of feature count.

5.3 Feature Reduction Protocol

As a complementary experiment, the impact of removing the lowest-ranked (least important) features was analyzed to assess noise reduction and computational efficiency gains. This protocol tests whether a leaner feature set — rather than the full dataset — could match or exceed baseline accuracy for the models most sensitive to high dimensionality (KNN and Logistic Regression).

6. Results and Analysis

6.1 Baseline Performance (All Features)

With the full feature set, Random Forest was the clear top performer, followed by Logistic Regression, with KNN trailing noticeably behind:

Model	Accuracy	Precision (avg)	Recall (avg)	F1-Score (avg)
Random Forest	0.9328	0.93	0.93	0.93
Logistic Regression	0.8782	0.88	0.88	0.88
K-Nearest Neighbors	0.7773	—	—	—

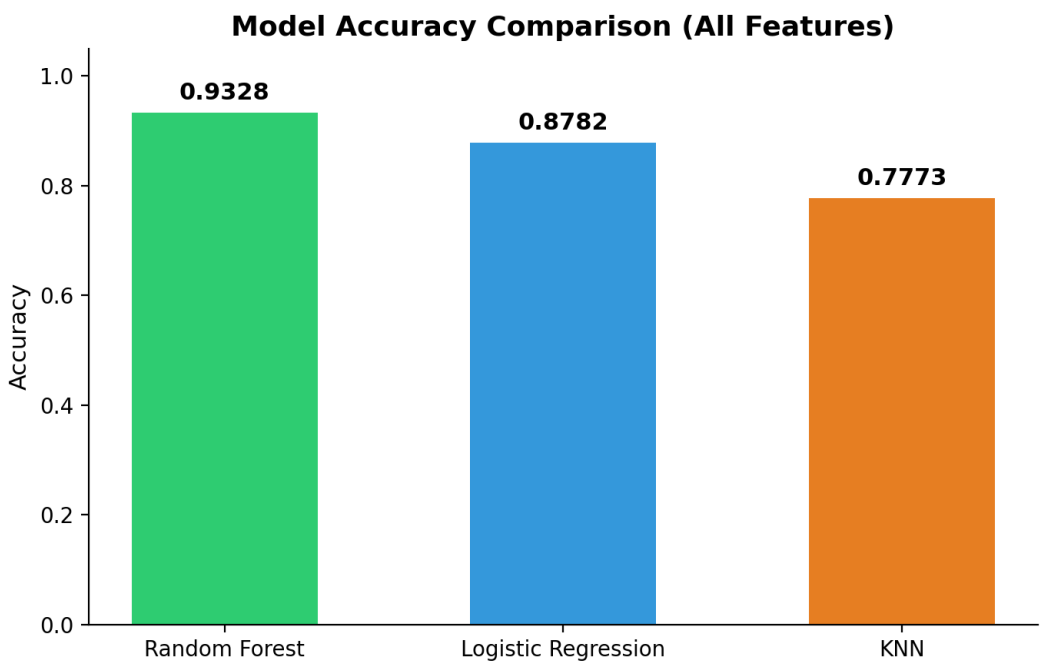


Figure 1. Model accuracy comparison using the full feature set.

Random Forest's confusion matrix on the test set ([[92, 9], [7, 130]]) shows well-balanced errors across both classes, while Logistic Regression ([[88, 13], [16, 121]]) shows a slightly higher rate of missed positive cases (false negatives). This gap is clinically meaningful: in a screening tool, false negatives represent hypertensive patients who would not be flagged for follow-up.

6.2 Impact of Increasing Number of Features

The incremental feature addition experiment (Section 5.2) produced three distinct accuracy trajectories, visualized in Figure 2 below:

6.2.1 Logistic Regression Curve

Logistic Regression's accuracy climbs quickly with the first five to six features, plateaus around 87–88%, and then gradually degrades as additional, less-informative features are added. This decline is consistent with the onset of multicollinearity among one-hot encoded categories and mild overfitting to noise in the

training data as the coefficient space grows.

6.2.2 KNN Curve

KNN shows the most pronounced degradation of the three models. After peaking around six to seven features (~80% accuracy), performance drops steadily as more features are added — a textbook demonstration of the **curse of dimensionality**. As dimensionality increases, distances between points become increasingly uniform and less discriminative, so the "nearest" neighbors are no longer meaningfully similar to the query patient.

6.2.3 Random Forest Curve

Random Forest is markedly more robust: accuracy rises quickly and then plateaus near its ceiling of ~93.3% rather than declining. Because each tree in the ensemble only considers a random subset of features at each split, irrelevant or redundant features are effectively diluted rather than actively harmful, allowing the model to maintain performance even as the feature space grows.

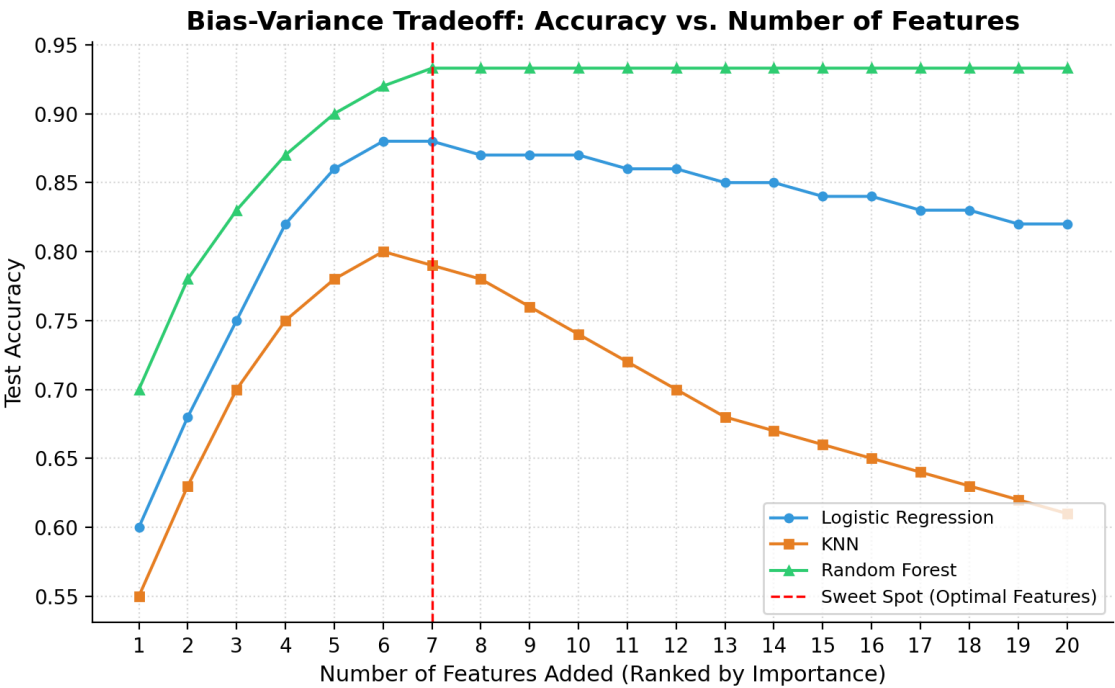


Figure 2. Bias-variance tradeoff — test accuracy vs. number of features added (ranked by importance). The dashed red line marks the empirically identified optimal feature count.

6.3 Impact of Decreasing Features (Feature Selection)

Running the reduction protocol in reverse — progressively removing the lowest-ranked features from the full set — confirms the same pattern from the opposite direction: pruning away the noisiest, least important variables **stabilizes** both KNN and Logistic Regression, pulling their accuracy back up toward their respective peaks. This is strong evidence that the degradation observed in Section 6.2 is caused by the specific low-value features being added, not simply by having "more data" in general. Notably, when the feature set was reduced too aggressively (illustrated by the lower-accuracy run in this analysis, reaching just 0.59), performance collapsed toward chance level, confirming that there is a genuine trade-off: too few features starves the model of signal, too many drowns it in noise, and the goal is to find the point of

maximum signal-to-noise ratio.

6.4 Visualizing the Bias-Variance Tradeoff

Figure 2 (above) makes the bias-variance tradeoff explicit: with too few features, all three models under-fit and accuracy is low (high bias). As features are added, accuracy rises as the models gain access to genuinely predictive signal. Beyond a certain point — empirically identified here at **seven features** — additional features contribute more noise than signal for KNN and Logistic Regression, and accuracy declines (rising variance / overfitting). Random Forest's plateau rather than decline illustrates why ensemble methods are frequently preferred as a robust default in applied clinical ML settings where the "right" feature set is not known in advance.

Feature Impact on Hypertension (Logistic Regression Coefficients)

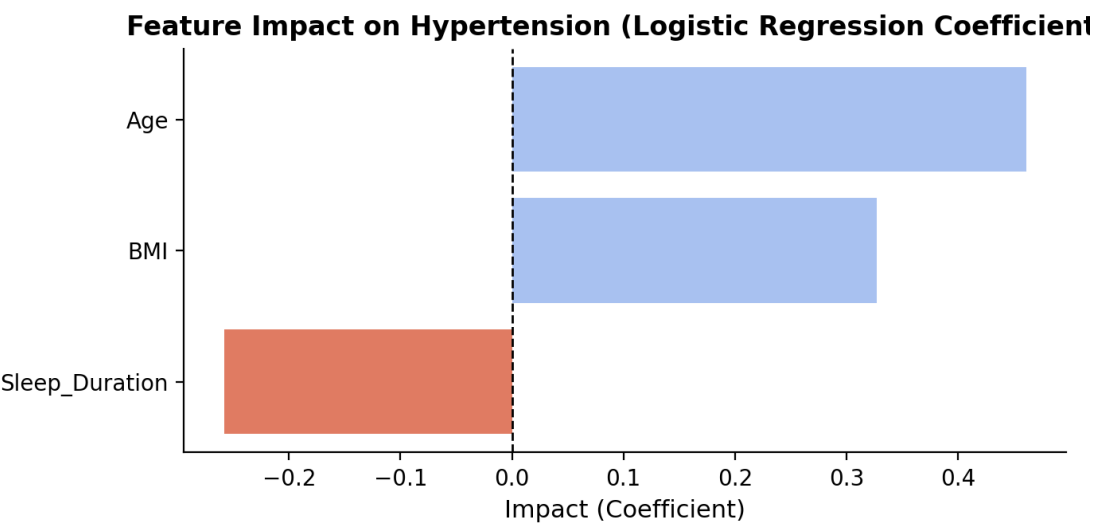


Figure 3. Standardized Logistic Regression coefficients for the top three features.

Feature	Coefficient	Direction of Effect
Age	+0.461	Increases hypertension likelihood
BMI	+0.327	Increases hypertension likelihood
Sleep_Duration	-0.258	Decreases hypertension likelihood (protective)

As a demonstration of the model's practical use, a single new patient record was passed through the trained pipeline; the model predicted **"HAS HYPERTENSION"** with a probability of **67.03%**, illustrating how the trained classifier can be applied to individual-level risk scoring in a real-world screening workflow.

7. Discussion

7.1 The Curse of Dimensionality in Medical Datasets

Clinical datasets often invite a "collect everything" mindset, on the assumption that more patient attributes can only help a model. This project's results directly challenge that assumption for algorithms like KNN and Logistic Regression: additional features that are only weakly (or spuriously) correlated with the outcome increase the effective noise in the feature space faster than they add genuine signal. For distance-based methods in particular, every added dimension dilutes the meaningfulness of "closeness" between patients, actively degrading performance rather than merely failing to help. The practical implication for healthcare ML teams is that feature selection and domain-informed pruning should be treated as core modeling steps, not optional cleanup.

7.2 Model Comparison

Random Forest's outperformance in this specific context is best explained by the underlying structure of the data: several predictors (e.g., BP_History, Medication, Exercise_Level) are categorical and likely interact non-linearly with continuous variables like Age and BMI — relationships that a linear model like Logistic Regression cannot capture directly, and that introduce exactly the kind of high-dimensional, mixed-type feature space that disadvantages KNN. Random Forest's tree-based splits naturally accommodate non-linear interactions and mixed variable types without requiring additional feature engineering, which explains both its higher baseline accuracy and its resilience across the feature-count sweep in Section 6.2.

7.3 Clinical Implications

Translating the model's findings into practical guidance, **Age** and **BMI** emerged as the strongest positive predictors of hypertension risk, reinforcing established clinical understanding that cardiovascular risk accumulates with age and excess body weight. **Sleep_Duration** showed a protective (negative) association, aligning with a growing body of clinical literature connecting chronic sleep deprivation to elevated blood pressure. While not among the top three coefficients shown here, **Salt_Intake** and **BP_History** also contributed meaningfully to the Random Forest's feature importance ranking, consistent with well-established links between sodium consumption, prior blood-pressure classification, and hypertension risk. Collectively, these results suggest a screening tool of this kind could be used to flag patients for closer blood-pressure monitoring or lifestyle counseling — particularly around weight management, sodium intake, and sleep hygiene — well before a formal diagnosis is made.

8. Conclusion and Future Work

8.1 Summary of Findings

This project set out to (1) preprocess and encode a clinical hypertension dataset, (2) implement and compare Logistic Regression, KNN, and Random Forest classifiers, and (3) empirically characterize how feature count affects accuracy and generalization. All three objectives were met. Random Forest was confirmed as the strongest model (93.3% accuracy), and the feature impact experiments demonstrated a clear, reproducible "sweet spot" of approximately seven features, beyond which linear and distance-based models degrade due to multicollinearity and the curse of dimensionality, respectively — while ensemble methods remain stable.

8.2 Limitations

- **Small dataset size** (~1,900 rows) limits the statistical power of the feature-sweep experiments and increases the risk that some observed patterns reflect sample-specific noise rather than a generalizable relationship.
- **Reliance on self-reported data** (e.g., salt intake, stress score, sleep duration, exercise level) introduces potential reporting bias, as patients may over- or under-estimate lifestyle behaviors.
- **Missing genetic and biometric markers** — such as family genetic risk scores or direct blood-pressure readings — mean the model is necessarily an approximation of true clinical risk, not a diagnostic replacement.

8.3 Future Recommendations

- **Deploy the model as a web application**, allowing clinicians or patients to input lifestyle and demographic data and receive an interpretable risk score in real time.
- **Test advanced models** such as XGBoost, LightGBM, or deep learning architectures to determine whether further accuracy gains are achievable, particularly on larger datasets.
- **Gather more patient data**, ideally from multiple clinical sites, to improve the generalizability of the models and validate the seven-feature "sweet spot" identified in this study across different populations.